MTH 200 Chapter 5: Integration

Summary of Integration Worksheet

Compute each of the following. Use algebra and u-substitution when appropriate.

1. $\displaystyle\int \frac{\tan^{-1}(x)}{1+x^2}\,dx$

   Let $u=\tan^{-1}(x)$, then $du=\dfrac{1}{1+x^2}dx$. Therefore, we have:

   $$\int \frac{\tan^{-1}(x)}{1+x^2}\,dx=\int u\,du=\frac{1}{2}u^2+C=\frac{1}{2}\left[\tan^{-1}(x)\right]^2+C$$

2. $\displaystyle\int_0^{\frac{\pi}{2}} \cos(x)\sin(\sin(x))\,dx$

   First we note that the function is continuous on $\left[0,\frac{\pi}{2}\right]$, so we can apply FTCP2.

   Let $u=\sin(x)$, then $du=\cos(x)dx$.

   Our limit of integration become: $u(0)=\sin(0)=0$ and $u\left(\frac{\pi}{2}\right)=\sin\left(\frac{\pi}{2}\right)=1$

   Therefore, we have:

   $$\int_0^{\frac{\pi}{2}} \cos(x)\sin(\sin(x))\,dx=\int_0^1 \sin(u)\,du=-\cos(u)\Big|_0^1=-\cos(1)+\cos(0)=1-\cos(1)$$

3. $\displaystyle\int \tan(x)\ln(\cos(x))\,dx$

   Let $u=\ln(\cos(x))$, then $du=\dfrac{1}{\cos(x)}\left[-\sin(x)\right]dx$ so $-du=\tan(x)dx$. Therefore, we have:

   $$\int \tan(x)\ln(\cos(x))\,dx=\int -u\,du=-\frac{1}{2}u^2+C=-\frac{1}{2}\left[\ln(\cos(x))\right]^2+C$$

4. $\displaystyle\int_0^{\frac{1}{4}} \frac{\sin^{-1}(2x)}{\sqrt{1-4x^2}}\,dx$

First we note that the function is continuous on $\left[0,\frac{1}{4}\right]$, so we can apply FTCP2.

Let $u=\sin^{-1}(2x)$, then $du=\dfrac{2}{\sqrt{1-(2x)^2}}dx$, so $\dfrac{1}{2}du=\dfrac{1}{\sqrt{1-4x^2}}dx$

Our limit of integration become: $u(0)=\sin^{-1}(0)=0$ and $u\left(\frac{1}{4}\right)=\sin^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{6}$

Therefore, we have:

$$\int_0^{\frac{1}{4}} \frac{\sin^{-1}(2x)}{\sqrt{1-4x^2}}\,dx=\int_0^{\frac{\pi}{6}}\frac{1}{2}u\,du=\frac{1}{4}u^2\Big|_0^{\frac{\pi}{6}}=\frac{1}{4}\left(\frac{\pi}{6}\right)^2=\frac{\pi^2}{144}$$

5. $\displaystyle\int \frac{t^3-4t^2+4t}{t^2-2t}\,dt$

Through simplification we see that:

$$\int \frac{t^3-4t^2+4t}{t^2-2t}\,dt=\int \frac{t\left(t^2-4t+4\right)}{t(t-2)}\,dt=\int \frac{t(t-2)(t-2)}{t(t-2)}\,dt=\int (t-2)\,dt=\frac{1}{2}t^2-2t+C$$

6. $\displaystyle\int_{\frac{-\pi}{6}}^{\frac{\pi}{6}} \tan^3(\theta)d\theta$

First we note that the function is continuous on $\left[-\frac{\pi}{6},\frac{\pi}{6}\right]$ and that $f(x)=\tan^3(x)$ is an odd function because: $f(-x)=\tan^3(-x)=\left[\tan(-x)\right]^3=\left[-\tan(x)\right]^3=-\tan^3(x)=-f(x)$.

Therefore, by properties of symmetry we know that:

$$\int_{\frac{-\pi}{6}}^{\frac{\pi}{6}} \tan^3(\theta)d\theta=0$$

7. $\int \frac{x^2}{x^2+1}dx$

$$\int \frac{x^2}{x^2+1}dx = \int \frac{x^2+1-1}{x^2+1}dx = \int \left[\frac{x^2+1}{x^2+1} - \frac{1}{x^2+1}\right]dx = \int \left[1 - \frac{1}{x^2+1}\right]dx = x - \tan^{-1}(x) + C$$

8. $\int_2^4 \frac{x^4-16}{x-4}dx$.

We note that $f(x) = \frac{x^2-16}{x-4}$ is <u>*NOT*</u> continuous on $[2,4]$, specifically because it is undefined at $x=4$. Therefore, the FTCP2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

9. $\int x^3\sqrt{x^2+1}dx$

Let $u = x^2+1$, then $du = 2x\,dx$ so $\frac{1}{2}du = x\,dx$. We also note that $u = x^2+1 \Leftrightarrow u-1 = x^2$.

Therefore, we have:

$$\int x^3\sqrt{x^2+1}dx = \int \frac{1}{2}(u-1)\sqrt{u}\,du = \frac{1}{2}\int\left(u^{3/2} - u^{1/2}\right)du = \frac{1}{2}\left(\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2}\right) + C = \frac{1}{5}\left(x^2+1\right)^{5/2} - \frac{1}{3}\left(x^2+1\right)^{3/2} + C$$

10. $\int \frac{2e^z+2}{e^z+z}dz$

Let $u = e^z + z$, then $du = \left(e^z+1\right)dz$. Therefore, we have:

$$\int \frac{2e^z+2}{e^z+z}dz = 2\int \frac{e^z+1}{e^z+z}dz = 2\int \frac{1}{u}du = 2\ln|u| + C = 2\ln\left|e^z+z\right| + C$$

11. $\int_0^1 \left(\sqrt[4]{x}+1\right)^2 dx$

First we note that the function is continuous on $[0,1]$, so we can apply FTCP2.
Therefore,

$$\int_0^1 \left(\sqrt[4]{x}+1\right)^2 dx = \int_0^1 \left(x^{1/4}+1\right)^2 dx = \int_0^1 \left(\left(x^{1/4}\right)^2 + 2x^{1/4}+1\right) dx = \int_0^1 \left(x^{1/2}+2x^{1/4}+1\right) dx$$

$$= \left(\frac{2}{3}x^{3/2}+\frac{8}{5}x^{5/4}+x\right)\Bigg|_0^1 = \left(\frac{2}{3}(1)^{3/2}+\frac{8}{5}(1)^{5/4}+1\right) - \left(\frac{2}{3}(0)^{3/2}+\frac{8}{5}(0)^{5/4}+0\right) = \left(\frac{2}{3}+\frac{8}{5}+1\right) - 0 = \frac{49}{15}$$

12. $\int \left[7+\cot^2(x)\right] dx$

$$\int \left[7+\cot^2(x)\right] dx = \int \left[7+\csc^2(x)-1\right] dx = \int \left[6+\csc^2(x)\right] dx = 6x-\cot(x)+C$$

13. $\int_{-1}^{0} \frac{x^2}{\sqrt{1-x}}\, dx$

First we note that the function is continuous on $[-1,0]$, so we can apply FTCP2.
Let $u=1-x$, then $du=-dx$, so $-du=dx$. We also note that $u=1-x \Leftrightarrow x=1-u \Leftrightarrow x^2=(1-u)^2$.
Our limit of integration become: $u(-1)=1-(-1)=2$ and $u(0)=1-0=1$
Therefore, we have:

$$\int_{-1}^{0} \frac{x^2}{\sqrt{1-x}}\, dx = \int_2^1 -\frac{(1-u)^2}{\sqrt{u}}\, du = \int_1^2 \left(1-2u+u^2\right)u^{-1/2} du = \int_1^2 \left(u^{-1/2}-2u^{1/2}+u^{3/2}\right) du = \left(2u^{1/2}-\frac{4}{3}u^{3/2}+\frac{2}{5}u^{5/2}\right)\Bigg|_1^2$$

$$= \left(2(2)^{1/2}-\frac{4}{3}(2)^{3/2}+\frac{2}{5}(2)^{5/2}\right) - \left(2(1)^{1/2}-\frac{4}{3}(1)^{3/2}+\frac{2}{5}(1)^{5/2}\right) = \left(2\sqrt{2}-\frac{4}{3}\sqrt{8}+\frac{2}{5}\sqrt{32}\right) - \left(2-\frac{4}{3}+\frac{2}{5}\right)$$

$$= \left(2\sqrt{2}-\frac{8}{3}\sqrt{2}+\frac{8}{5}\sqrt{2}\right) - \left(\frac{16}{15}\right) = \frac{14}{15}\sqrt{2}-\frac{16}{15} = \frac{2}{15}\left(7\sqrt{2}-8\right)$$

14. $\displaystyle\int \frac{x^5+x^2}{1+x^6}\,dx$

First we simplify by writing this as a sum of two integrals.

$$\int \frac{x^5+x^2}{1+x^6}\,dx = \int\left[\frac{x^5}{1+x^6}+\frac{x^2}{1+x^6}\right]dx = \int \frac{x^5}{1+x^6}\,dx + \int \frac{x^2}{1+x^6}\,dx$$

Integral #1:

Let $u=1+x^6$ and thus $\dfrac{du}{dx}=6x^5 \quad\to\quad du=6x^5dx \quad\to\quad \dfrac{1}{6}du=x^5dx$.

Integral #2:

Let $v=x^3$ and thus $\dfrac{dv}{dx}=3x^2 \quad\to\quad dv=3x^2dx \quad\to\quad \dfrac{1}{3}dv=x^2dx$

$$\int \frac{x^5+x^2}{1+x^6}\,dx = \int \frac{x^5}{1+x^6}\,dx + \int \frac{x^2}{1+x^6}\,dx = \frac{1}{6}\int \frac{1}{u}\,du + \frac{1}{3}\int \frac{1}{1+v^2}\,dv = \frac{1}{6}\ln|u| + \frac{1}{3}\tan^{-1}(v)+C$$

$$= \frac{1}{6}\ln\left|1+x^6\right| + \frac{1}{3}\tan^{-1}\left(x^3\right)+C$$

15. $\displaystyle\int \frac{3x^8}{x^3+4}\,dx$

Let $u=x^3+4$ and thus $du=3x^2dx$

If we make the replacement right now then we will still have an x^6 present. Notice that if

$$u=x^3+4 \quad\to\quad u-4=x^3 \quad\to\quad (u-4)^2=x^6$$

$$\int \frac{3x^8}{x^3+4}\,dx = \int \frac{3x^2\cdot x^6}{x^3+4}\,dx = \int \frac{(u-4)^2}{u}\,du = \int \frac{u^2-8u+16}{u}\,du = \int\left(u-8+\frac{16}{u}\right)du$$

$$= \frac{u^2}{2}-8u+16\ln|u|+C = \frac{1}{2}\left(x^3+4\right)^2-8\left(x^3+4\right)+16\ln\left|x^3+4\right|+C$$

16. $\displaystyle\int_{-1}^{4} \frac{2x^2+7x^5}{x^4}\,dx$

We note that $f(x)=\dfrac{2x^2+7x^5}{x^4}$ is *NOT* continuous on $[-1,4]$, specifically because it is undefined at $x=0$. Therefore, the FTCP2 does not apply and we *currently* have no way to determine such an area. We cannot proceed.

17. $\int e^{5x}dx$

Let $u = 5x$ and thus $du = 5dx \quad \rightarrow \quad \frac{1}{5}du = dx$

$$\int e^{5x}dx = \int \frac{1}{5}e^{u}du = \frac{1}{5}e^{u} + C = \frac{1}{5}e^{5x} + C$$

18. $\int x\sqrt{x^2-1}\,dx$

Let $u = x^2 - 1$ and thus $du = 2xdx \quad \rightarrow \quad \frac{1}{2}du = xdx$

$$\int x\sqrt{x^2-1}\,dx = \int \frac{1}{2}\sqrt{u}du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}\left(x^2-1\right)^{3/2} + C$$

19. $\displaystyle\int_1^4 \frac{4x^3 - \sqrt{x}e^x - 3\sin\left(\sqrt{x}\right)}{3\sqrt{x}}dx$

First we note that the function is continuous on $[1,4]$, so we can apply FTCP2.

We note that:

$$\int_1^4 \frac{4x^3 - \sqrt{x}e^x - 3\sin\left(\sqrt{x}\right)}{3\sqrt{x}}dx = \int_1^4 \left[\frac{4}{3}x^{5/2} - \frac{1}{3}e^x - \frac{\sin\left(\sqrt{x}\right)}{\sqrt{x}}\right]dx = \int_1^4 \left[\frac{4}{3}x^{5/2} - \frac{1}{3}e^x\right]dx - \int_1^4 \frac{\sin\left(\sqrt{x}\right)}{\sqrt{x}}dx$$

The second integral here will require a u-substitution.

Let $u = \sqrt{x}$, then $du = \frac{1}{2\sqrt{x}}dx \quad \Rightarrow \quad 2du = \frac{1}{\sqrt{x}}dx$

Our limits of integration become: $u(1) = \sqrt{1} = 1$ and $u(4) = \sqrt{4} = 2$

Therefore, we have that:

$$\int_1^4 \left[\frac{4}{3}x^{5/2} - \frac{1}{3}e^x\right]dx - \int_1^4 \frac{\sin\left(\sqrt{x}\right)}{\sqrt{x}}dx = \int_1^4 \left[\frac{4}{3}x^{5/2} - \frac{1}{3}e^x\right]dx - \int_1^2 \sin(u)\cdot 2du$$

$$= \left[\frac{4}{3}\cdot\frac{2}{7}x^{7/2} - \frac{1}{3}e^x\right]\Bigg|_1^4 + 2\cos(u)\Big|_1^2$$

$$= \left[\frac{8}{21}(4)^{7/2} - \frac{1}{3}e^4\right] - \left[\frac{8}{21}(1)^{7/2} - \frac{1}{3}e^1\right]\left[2\cos(2)\right] - \left[2\cos(1)\right]$$

$$= \frac{1016}{21} - \frac{1}{3}e^4 + \frac{1}{3}e + 2\cos(2) - 2\cos(1)$$

$$= \frac{1016}{21} - \frac{1}{3}\left(e^4 - e\right) + 2\left[\cos(2) - \cos(1)\right]$$